

Week 3: NumPy and Pandas

Objective

To learn the fundamentals of data analysis using NumPy and Pandas and perform data cleaning operations on datasets.

Concepts Learned

- Introduction to NumPy
- Arrays and Array Operations
- Introduction to Pandas
- Series and DataFrames
- Handling Missing Values
- Data Cleaning and Analysis

Project: Dataset Cleaning and Analysis

Developed a dataset cleaning and analysis program using Pandas. The program handled missing values, organized data into a DataFrame, and calculated statistical information such as average marks.

Outcome

Successfully learned how to work with datasets using Pandas and NumPy. Gained practical experience in cleaning data, handling missing values, and performing basic data analysis.

Tools Used

- Python
- NumPy
- Pandas
- Visual Studio Code (VS Code)

Conclusion

Week 3 provided hands-on experience in data analysis and preprocessing. The project demonstrated how Python libraries such as NumPy and Pandas can be used to clean, organize, and analyze datasets efficiently, which is essential for data science and machine learning applications.